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23 March 1984
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9 July 1971
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15 March 1974

MILITARY SPECIFICATION

GYROSCOPE ASSEMBLIES AND ATTITUDE AND DIRECTIONAL REFERENCE INSTRUMENTS FOR AIRCRAFT; PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers procedures for the packaging of sensitive gyroscope assemblies and other attitude and directional reference instruments for aircraft (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-E-485	Enamel, Semi-gloss, Rust-inhibiting
TT-E-529	Enamel, Alkyd, Semi-gloss

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Code 9321, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

NO DELIVERABLE DATA REQUIRED BY THIS DOCUMENT

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SPECIFICATIONS (continued)

FEDERAL (continued)

TT-P-1757	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-60	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3133	Cellular Elastomeric Materials, Molded or Fabricated Parts
MIL-C-5501	Cap and Plug, Protective, Dust and Moisture Seal
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
DoD-P-15328	Primer (Wash), Pretreatment
MIL-P-15930	Primer Coating, Shipboard, Vinyl-Zinc Chromate
MIL-P-23377	Primer Coating; Epoxy Polyamide, Chemical and Solvent Resistant
MIL-V-27166	Valve, Pressure Equalizing, Gaseous Products
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-P-52192	Primer Coating, Epoxy
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-101	Test Procedures for Packaging Materials
FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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STANDARDS (Continued)

MILITARY (Continued)

MIL-STD-670	Classification System and Tests for Cellular Elastomeric Materials
MIL-STD-794	Parts and Equipment, Procedures for Packaging of
MS25043	Cover, Electrical Connector, Receptacle, AN Type
MS90376	Cap, Dust, Plastic, Electric Connector

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D951	Water Resistance of Shipping Containers by Spray Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL AEROSPACE STANDARDS

NAS 813	Cap - Protective, Electrical Connector
NAS 820	Plug - Protective, Electrical Connector

(Application for copies should be addressed to the Aerospace Industries Association, 1725 DeSales Street, N.W., Washington, DC 20036.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to a first article inspection (see 4.3, 6.2 and 6.3).

3.2 Definition of terms. For purposes of this specification, the following definitions shall apply:

a. Container system - the shipping container only with a dummy load installed
b. Pack - the shipping container including the applicable item packed as for shipment

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3.3 Packaging requirements.

3.3.1 General. Packaging of gyroscope assemblies or other attitude reference display instruments shall meet the requirements for level A or B protection of MIL-STD-794, as specified (see 6.2). The method of cleaning shall be C-1 of MIL-P-116. The method of preservation shall be IC, IA or II, as specified (see 6.2), accomplished in accordance with the detail requirements specified herein. The gyroscope assembly or other attitude reference display instrument shall be packaged in shipping containers prescribed herein. The interior handling case, when used, shall be designed so that its insertion and removal from the exterior container can be easily accomplished.

3.3.2 Detail. When exposed terminals or connectors subject to corrosion are present, the terminals or connectors shall be closed with metal caps or plugs conforming to MIL-C-5501 or MS25043, plastic caps or plugs conforming to MS90376, NAS 813 or NAS 820, or waterproof tape conforming to PPP-T-60, or equivalent. The complete unit, after cushioning any sharp edges or points with nondusting, noncorrosive cushioning material as necessary, insertion of desiccant in accordance with MIL-P-116, if required, and enclosing in a heat-sealed bag fabricated of appropriate material, shall, when specified, be immobilized in an interior handling case (see 6.2). After closing the interior handling case, it shall be centrally placed in the shock-isolation media of the exterior container. The exterior container shall be closed so as to accomplish a water resistant pack.

3.3.3 New materials. The use of newly developed packaging materials or products is encouraged and recommended providing they are equal or better than similar approved materials or products without increasing the overall cost to the Government. To expedite their use prior to their inclusion in a Government specification, their use will be permitted under conditions outlined herein. In instances where the material or product is not covered by a specific military specification, the manufacturer or fabricator of the material or product shall furnish documented evidence certified by an approved testing laboratory, that the material or product meets or exceeds all requirements of the performance specification for a similar material or product. The request for approval shall be submitted to the acquiring activity via the technical packaging element of the contract administration activity. If, after a review of the material or product and related certified compliance report, or the witnessing of the stipulated tests, it is the opinion of the approving authority that the material or products meets or exceeds the criteria established for similar material or products, interim authorization for use may be granted pending preparation of a new specification.

3.4 General design requirements. The shipping container shall be designed to accomodate the gyroscope assembly or other attitude reference instrument through its entire logistical cycle, which includes initial delivery of instrument from the supplier through retrograde shipments. The container with its interior preservation and packaging provisions shall protect the equipment from damaging shock, vibration, rough handling, or contamination that might occur during handling, transit and storage (see 6.1). The packaged equipment shall be quickly and easily removable from the container with the use of simple tools. The container shall function satisfactorily in ambient temperatures ranging from -20° to 120°F.

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3.5 Detail design requirements.

3.5.1 Interior handling case. If an interior handling case is used (see 6.2), it shall be molded of fiberglass, virgin polyethylene, or other virgin synthetic/plastic materials, or fabricated from synthetic/plastic foams, plastic board, corrosion resistant metals or a combination of these materials. Weather-resistant corrugated fiberboard boxes conforming to PPP-B-636 may be used, if specified (see 6.2). Unless corrugated materials are used a handle shall be provided. Closure hardware shall be captive. The unit to be packaged shall be centrally located and snugly restrained within the handling case using convoluted polyurethane or other suitable cushioning material having a minimum thickness of one inch. The size of the handling case cavity shall be as specified by the acquiring activity (see 6.2).

3.5.2 Exterior container. Unless otherwise specified the container shall be molded of fiberglass, virgin polyethylene, or other virgin synthetic/plastic materials, or fabricated from synthetic/plastic foams, plastic board, or corrosion resistant metal or a combination of these materials. Containers so constructed shall consist basically of a lower and upper shell (body and cover). The mating surfaces of the upper and lower shells shall be secured with recessed captive quarter-turn fasteners or overcenter spring loaded latches. A water resistant seal shall be maintained by a suitable tongue-in-groove type closure with the groove portion located in the container cover. The gasket shall be one continuous length and easily replaceable. Separately applied sealing compounds or materials shall not be allowed. When assembled, the loading force on the gasket shall be uniform and no greater than that required for sealing. Weather-resistant corrugated fiberboard boxes conforming to PPP-B-636 or wood containers may be employed, when specified (see 6.2).

3.5.2.1 Size and equipment weight. The overall size of the exterior container shall be engineered to minimum dimensions consistent with the exterior dimensions of the item or the interior handling case, as applicable, the weight range of items being packaged, and the shock attenuation and vibration isolation required to meet the rough handling tests specified in 3.7.1 (see 6.2).

3.5.2.2 Cover. Unless otherwise specified (see 6.2), exterior container covers shall be detachable from the base. The cover shall be so designed to prevent damage to the attaching hardware during impacts as encountered in service and testing. Also, provisions shall be incorporated to ensure a unique orientation of the detachable cover when mated with the base of the container.

3.5.2.3 Handles. Unless otherwise specified (see 6.2), handles shall be provided on all containers other than those constructed of wood or corrugated fiberboard. Handles shall be a chest type designed for surface mounting with stops to hold the bail at a right angle to the mounting plate when in a carrying position and springloaded to fold the bail into a down position when released. The bail of the handle shall have an inside length of not less than 4.25 inches with a minimum clearance of 1.8 inches between the bail grip and the mounting plate. Handles shall be attached by sealed rivets. The handles of the container shall not malfunction nor shall the handles unduly deform or cause the container to deform when the container is tested in accordance with 4.5.2. Two handles (one on each end) will be

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required for containers with an anticipated gross weight under 120 pounds; four handles (two on each end) for containers with an anticipated gross weight over 120 pounds.

3.5.2.4 Protection of external hardware. The container shall incorporate bosses, corrugations or recesses so as to provide protection for any exterior hardware and handles.

3.5.2.5 Shock and vibration isolation. The container shall have provisions for isolating the item or interior handling case, as applicable, from shock and vibration encountered in shipment and storage. This may be accomplished by use of a shock-mounted platform securely attached to the inside of the exterior container or a resilient cushioning system. The shock isolation system shall be capable of satisfactory operation over the temperature range specified in 3.4.

3.5.2.6 Platform. If a shock mounted platform is employed (see 3.5.2.5), the platform shall either be molded of the same material as the container shell or fabricated from aluminum. Platform size shall be sufficient to accomodate the applicable interior handling case with item installed. The platform shall have a minimum one inch vertical flange around its outer edge. A 1/4 inch thick, open cell, neoprene sponge mat conforming to MIL-C-3133 and MIL-STD-670, Type 5, Class SC, Grade SCO-3 (Groendyk #571-N or equivalent) shall be provided on the mounting surface of the platform. Aluminum, if used, shall be anodized in accordance with MIL-A-8625. The payload shall be secured to the platform with two or three tie-down straps (AGM Cargo Ties, P/N (08992)CT1460 or equivalent). Two straps shall traverse the width of the platform. The third strap, when required, shall be located along the longitudinal centerline of the platform. Straps shall be captive.

3.5.2.7 Pressure relief valve. Unless otherwise specified (see 6.2), an automatic pressure/vacuum relief valve with manual pressure release and minimum flow rate conforming to MIL-V-27166 shall be installed in the exterior container wall for all containers other than those constructed of wood or corrugated fiberboard. The valve shall crack at a pressure differential of 1.0 psig (± 0.15) and shall reseal at a minimum pressure differential of 0.5 psig for both vacuum and pressure relief. Cracking and resealing pressure may be lowered to 0.5 psig (±0.1) and .25 psig respectively, if the design of the container inherently allows expansion and contraction of the container walls to compensate for this decreased pressure differential. This reduction, however, must be demonstrated by appropriate pressure/temperature testing by the container manufacturer.

3.5.2.8 Accessory attachment. A commercially acceptable waterproof seal (i.e. gasket) shall be required for all externally mounted accessories.

3.5.2.9 Fork lift provisions. Containers over 40 inches in the longest dimension or over 150 pounds gross weight shall be provided with appropriately sized fork lift openings or skids.

3.5.3 Color and surface treatment. The color of the exterior container, and the interior handling case if present, shall be specified by the acquiring agency (see 6.2), and shall be selected from FED-STD-595.

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Surface preparation and painting, when required, shall be accomplished for specific materials of construction as follows:

a. Plastic/Fiberglass. The color shall be obtained by means of pigmentation in the plastic or molded in gel coat.

b. Aluminum. When there is no painting required by the procuring agency the aluminum shall be cleaned and finished in accordance with MIL-C-5541 Type 3. When there is a painting requirement the following shall apply:

Pretreatment:

1. Chromate Conversion Coating as per MIL-C-5541
OR
2. Wash Primer as per DoD-P-15328

Primer (one coat):

1. Alkyd Zinc Chromate as per TT-P-1757
OR
2. Polyamide Epoxy as per MIL-P-23377

Topcoat (two coats):

1. Alkyd Enamel as per TT-E-529
OR
2. Polyurethane as per MIL-C-83286 or MIL-C-46168

c. Steel

Pretreatment:

1. Phosphated as per TT-C-490
OR
2. Sandblasted
OR
3. Wash Primer as per DoD-P-15328

Primer (one coat):

1. Epoxy Chromate as per MIL-P-52192
OR
2. Modified Alkyd as per TT-E-485 (second coat may also be used for topcoat)
OR
3. Vinyl Zinc Chromate as per MIL-P-15930

Topcoat:

1. Alkyd Enamel as per TT-E-529 (two coats)
OR
2. Polyurethane as per MIL-C-83286 (one coat) or MIL-C-46168 (one coat)

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3.6 Marking.

3.6.1 Marking for identification. All major components of the shipping container which are replaceable shall be properly identified in accordance with MIL-STD-130.

3.6.2 Precautionary markings. The following precautionary markings shall appear on two opposite sides of the interior handling case, when used, and exterior containers.

DELICATE INSTRUMENT
HANDLE WITH EXTREME CARE

In addition, both interior cases and exterior containers shall be marked "UP" with arrows on two opposite sides to indicate proper orientation. Interior handling cases shall also be marked "HANDLING CASE ONLY - DO NOT USE AS SHIPPING CONTAINER." Letters shall be at least two inches high. In cases where the container size precludes letters of this height, letters shall be as large as compatible with the existing space.

3.6.3 Special markings. Interior cases, when used, and exterior containers shall be marked "RETAIN FOR RETURN SHIPMENT OR RE-USE" in letters at least one inch high, and on two opposite sides of each container. Where size of containers precludes one inch high letters, the size shall be as large as compatible with the existing space.

3.6.4 Container marking for shipment and storage. Marking for shipment and storage shall be in accordance with MIL-STD-129. Marking of single application exterior containers shall be accomplished either by label or tag. Marking of multi-application exterior containers shall be accomplished by tag only. Bar code marking shall not be required for multi-application reusable containers.

3.7 Performance requirements.

3.7.1 Rough handling. The shipping container shall be capable of functioning satisfactorily after subjecting a container system to the rough handling tests (4.5.1). The container and all essential accessories shall reveal no significant structural weaknesses. No deformation/damage shall occur that will restrict dissassembly, reassembly and subsequent reuse of the container. Except for corrugated containers, any permanent deformation in the container shell or any integral part in excess of 0.375 inches beyond its normal plane or position shall be considered a significant structural weakness. In addition, the shock and vibration levels transmitted to the contents of the container shall not exceed the allowable shock and vibration levels. There shall be no evidence of shifting of the contents within the container that would create conditions likely to cause damage during shipment.

3.7.1.1 Drop Test. Unless otherwise specified (see 6.2), there shall be no maximum resultant peak acceleration readings over 15G's transmitted to the specified measuring point of the dummy load (see 4.3.3) when tested in accordance with paragraph 4.5.1.1. No evidence of isolator bottoming shall be allowable on accelerometer output signal records even if the 15G limit is not exceeded.

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.3.7.1.2 Vibration transmissibity and fatigue. Unless otherwise specified (see 6.2), the product of the transmissibility to the dummy load at any frequency times the expected worst case input acceleration from various transportation sources as defined in Fig. 1 shall not exceed the 15G fragility requirement of the item. The transmissibility shall be determined by conducting the vibration test of 4.5.1.2. Additionally, at the conclusion of the vibration test there shall be no evidence of shifting of contents or other permanent degradation of the container that could cause subsequent damage during shipment.

3.8 Water resistance. The exterior container shall not leak when tested in accordance with 4.5.3. Corrugated containers are excluded from this test.

3.9 Stacking. The exterior container shall not suffer permanent deformation, cracking or other detrimental effects when tested in accordance with 4.5.4.

3.10 Form and fit. The actual item for which the container is designed shall fit properly in the container when tested in accordance with 4.5.5.

Regions of vibration occurrence in various transportation media

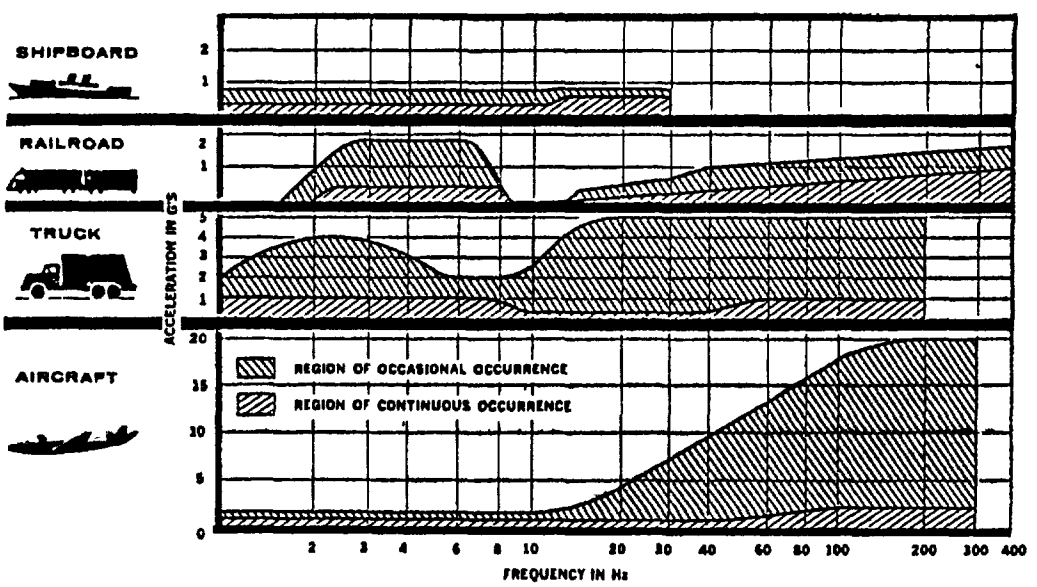


Figure 1. Transportation vibration

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3.11 Handle performance. Container with handles shall show no deformation or damage to the handle or container following the handle test specified in 4.5.2.

3.12 Workmanship. Workmanship shall be in accordance with the best commercial practice. Defects, imperfections or omissions which would impair the effectiveness of the complete pack or any parts thereof shall not be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classsification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection (see 6.3).

4.3.1 Notification. The contractor shall notify the acquiring activity in sufficient time in advance of first article inspection to permit Government representation at the test site.

4.3.2 First article inspection samples. The first article inspection sample shall consist of one complete pack or container system, as applicable, representative of production items. Detail manufacturing drawings of the container shall be available for review by the Government representative prior to testing.

4.3.3 First article examination and tests. First article inspection shall consist of the visual examination and all tests specified herein. Visual examination shall be in accordance with 4.4.2. Testing shall be in accordance with 4.5 and, except for the form and fit test (4.5.5), shall be conducted using a dummy load to simulate the actual item being packaged. The dummy load shall be representative of the item being packaged in size, configuration, weight, center of gravity and mounting points. If internal vibration isolation is an integral part of the item's design, the dummy load shall include an identical isolation system in its design. The dummy load shall be instrumented with a triaxial accelerometer mounted as near as practicable to the center of gravity to record peak acceleration simultaneously in three mutually perpendicular directions. The response of the complete measuring system, from accelerometer through the readout instrument, shall be a minimum of ninety percent over the frequency range with

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a lower limit of 2Hz and an upper limit equal to 15 times the resonant frequency of the shock isolation system. Filtering shall be used to limit the system response to this frequency range.

4.3.3.1 Sequence of testing. Sequence of testing shall be drop test, vibration test, handle test, water resistance test, stacking test, and form and fit test, as applicable.

4.3.4 Approval. All first article tests shall be supervised and witnessed by a Government representative at the test location. Authorization to utilize the pack on a production basis is contingent on satisfactory completion of the first article inspection and test and subsequent approval by the witnessing Government representative.

4.3.5 Results. First article test results, including method of instrumentation for same, shall be available for review by the Government representative after tests are completed.

4.4 Quality conformance inspection. Sample packs shall be selected and inspected in accordance with MIL-STD-105, Inspection Level S-4, Acceptable Quality Level 1.5 percent defective. The unit of sampling shall be one complete pack. For packs, quality conformance inspection shall consist of the visual examination (see 4.4.2) only. For containers, quality conformance inspection shall be conducted on only one sample container, regardless of lot size. Inspection shall include the visual examination (see 4.4.2), instrumented drop test (see 4.5.1.1) at room temperature only using a dummy load as defined in 4.3.3, and the water resistance test, if applicable (see 4.5.3). No defects shall be allowable during quality conformance inspection of containers.

4.4.1 Lot. A lot shall consist of all packs or containers made of the same materials during an identifiable period, and submitted at one time for inspection.

4.4.2 Visual examination. A thorough visual examination shall be conducted with emphasis on the following points, as applicable:

a. Conformance of container to the approved drawing definition of same.

b. Finish (for burrs, sharp edges, delaminations, etc.), if applicable.

c. Handle operation, size, location and clearance, if applicable.

d. Shock isolation system stability.

e. Correct alignment of cover attaching hardware with corresponding hardware on container bottom, if applicable.

f. Conformance of all marking, as required.

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4.5 Tests.

4.5.1 Rough handling tests.

4.5.1.1 Drop test. Drop tests shall be conducted on container systems. The drop test shall consist of the free-fall flat, edge, and corner drops prescribed in Procedure A, Test Method 5007, of FED-STD-101. Drop heights shall be those listed for the level of packaging required (see 3.2 and 6.2). A series of instrumented tests shall be conducted at room temperature and at -20°F. Container systems with gross weight over 150 pounds shall be subjected only to the rotational drop tests and impact tests specified in MIL-STD-794.

4.5.1.2 Vibration test. The vibration test shall be performed in the following manner: The container system, in its normal shipping position, shall be rigidly attached to a vibration exciter. Instrumentation shall be used to obtain transmissibility data. Transmissibility shall be determined by the ratio of the maximum resultant acceleration (vector summation of the accelerations received on the three major axes at any instant transmitted to the dummy load) divided by the input acceleration of the vibration exciter. A search for resonance shall be conducted by applying sinusoidal vibration in the vertical direction. Transmissibility data shall be obtained for the fundamental translational vibration mode over a frequency range of five (5) to fifty (50) hz. Input vibration shall be 1.0g or less. Sweep rate shall be approximately logarithmic and the total sweep time shall be five (5) minutes. A dwell test of 30 minutes duration (the test may be interrupted, if necessary, to prevent excessive temperature rise of resilient materials) shall be conducted at the major resonant frequency. The input of excitation shall be 0.125 inch double amplitude or 1.0g, whichever is the lesser value. Adjustments in input frequency may be necessary during the course of the test to compensate for changes in resonant frequency. The test shall be performed at ambient temperature (70° ± 20°F) unless otherwise specified. The temperature of resilient materials should not be allowed to exceed the safe operating temperature of the elastomer. When non-skidded fiberboard or wood containers are utilized, the test shall be repeated with vibration excitation applied in the transverse and longitudinal directions with the container in its normal shipping position.

4.5.2 Handle test. When applicable, the container system shall be suspended so that it swings free above the ground by fastening a rope through one handle. Test duration shall be one minute (+1, -0). This test shall be repeated on each handle individually.

4.5.3 Water resistance test. When applicable, the container system shall be tested in accordance with ASTM Test Method D951. The water spray shall be directed at all surfaces of the container for a minimum of ten minutes on each surface. The container shall then be examined for water penetration. Evidence of leakage shall be cause for rejection.

4.5.4 Stacking test. A weight equal to three times the gross shipping weight of the container system shall be placed on the container system at ambient temperature (70° ± 20°F) in such a manner as to simulate the effects of stacking similar containers on the test container. The weight

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shall remain in this position for 16 hours. At the end of this time, the container system shall be examined for permanent deformation, cracking or other detrimental effects.

4.5.5 Form and fit test. A pack shall be prepared by installing the actual item for which the container is designed in accordance with the installation instructions. The container shall be inspected for proper fit of the item in the container. In addition, the interior handling case installation instructions, when required, shall be verified for adequacy, clarity, completeness and proper location. Satisfactory item removal procedures shall also be verified.

5. PACKAGING

5.1 Not applicable.

6. NOTES

6.1 Intended use. The requirements specified herein are intended to ensure satisfactory procurement and use of shipping containers that will assure proper and safe handling, shipment, and storage of gyroscope assemblies and other attitude or directional reference instruments. The complete logistic cycle covers shipments to Government activities, shipments between storage depots, to and from overhaul and repair shops, and shipment to industry for rework and return to the Government. The procedures outlined herein are intended to prevent internal damage to the item and to optimize the duration of satisfactory service performance. The optional interior handling case is intended as a carrying case only for use immediately prior to installation of the contents in an assembly. The interior handling case alone does not provide adequate shock protection for shipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. When first article inspection is required (see 3.1 and 6.3).

c. Level of packaging required (see 3.3.1).

d. Method of preservation required (see 3.3.1).

e. When interior handling case is required and its cavity size when used (see 3.5.1).

f. If fiberboard containers are acceptable as interior handling cases and/or exterior containers (see 3.5.1 and 3.5.2).

g. Weight range of items being packaged (see 3.5.2.1).

h. If detachable exterior container cover is not required (3.5.2.2).

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i. If handles are not required on exterior container (3.5.2.3).

j. If pressure relief valves are not required (3.5.2.7).

k. Color of exterior container and interior handling case, as applicable (3.5.3).

l. Other fragility limits for items being packaged, when appropriate (see 3.7.1.1 and 3.7.1.2).

6.3 First article. When first article inspection is required, the contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article. First article testing shall be performed when one of the following conditions applies:

a. The container or pack is being fabricated by a manufacturer who has not previously submitted a first article sample.
b. The container material construction or shock mitigating system has changed since the previous first article submission.
c. The testing requirements have changed since the previous first article submission.

When the design adequacy of the container or pack has been previously verified or when a contractor is in continuous production of the pack from contract to contract, consideration should be given to waive the first article inspection. First article approval or waiving of first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 69

Review activities:
Army - SM
Air Force - 99

User activities:
Air Force - 71

Preparing activity:
Navy - AS
(Project No. PACK-0568)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-G-81559C	Gyroscope Assemblies and Attitude and Directional Reference Instruments For Aircraft; Packaging Of

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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